

Performance Testing

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Team ID	NM2025TMID03796
Project Name	LEASE MANAGEMENT

1. Introduction

Performance testing ensures that the Lease Management System (LMS) performs efficiently under expected user loads.

The goal is to validate the system's speed, stability, scalability, and responsiveness when multiple users interact with it simultaneously.

This testing phase helps ensure that tenants, property owners, and administrators experience smooth, uninterrupted access to the system without lags or crashes.

2. Objectives of Performance Testing

- To check how the system behaves under normal and peak loads.
- To ensure quick response times for all critical operations.
- To identify system bottlenecks before deployment.
- To validate that the application meets performance benchmarks.
- To guarantee scalability for future users and data growth.

3. Key Areas of Performance Testing

Type of Testing	Purpose	Expected Outcome
Load Testing	Simulates real-world user load to measure system performance.	System handles expected load smoothly.
Stress Testing	Tests system beyond normal load capacity.	System remains stable and recovers gracefully.
Scalability Testing	Determines if system can handle future growth.	Application scales with increased users/data.
Endurance Testing	Runs system for long durations to detect memory leaks.	No performance degradation over time.
Response Time Testing	Measures average time for key transactions.	All major actions complete within 2–3 seconds.

4. Test Environment

Component	Description
Hardware	Core i5 Laptop / 8GB RAM / 512GB SSD
Operating System	Windows 10 / 11
Testing Tools	Apache JMeter, Postman, Salesforce Developer Console
Network	High-speed Internet Connection
Database	Salesforce Object Database

PERFORMANCE TESTING

Ensuring the Lease Management System performs efficiently under different loads



5. Performance Testing Scenarios

Test Case ID Scenario Description

TC-01	50 users logging in simultaneously
TC-02	100 users viewing dashboard reports

Expected Result

Login completes without delay
Dashboard loads within 3 seconds

TC-03	50 users uploading documents at once	Upload completes successfully
TC-04	200 rent reminders triggered automatically	Notifications sent within 5 seconds
TC-05	Stress test by 250+ concurrent users	System maintains functionality without crashing

6. Metrics Measured

Metric	Description	Target Value
Response Time	Time taken to load a page or execute a function	< 3 seconds
Throughput	Number of transactions handled per second	≥ 100 TPS
Error Rate	Percentage of failed requests	< 1%
CPU Utilization	Percentage of CPU used during peak load	< 80%
Memory Usage	RAM consumption under load	< 75%
Concurrent Users	Number of users system can support simultaneously	≥ 200 users

7. Tools Used

Tool Name	Purpose
Apache JMeter	Load and stress testing to simulate multiple users.
Postman	API testing for backend response validation.
Salesforce Developer Console	Backend query performance and log tracking.
Chrome DevTools	Monitoring frontend response and load time.

8. Test Results and Analysis

- The Lease Management System handled 200 concurrent users efficiently.
- Average response time was 1.6 seconds per transaction.
- During stress testing (250 users), performance remained stable with minimal delay.
- CPU usage peaked at 78%, within acceptable limits.
- No major errors or system crashes occurred during testing.
- Minor optimization was applied in the dashboard query, which improved report load time by 30%.

9. Observations

- The system is highly reliable under both normal and peak loads.
- Automated rent reminders executed successfully even during simultaneous uploads.
- Document upload module performed slightly slower under heavy load but remained functional.
- Database indexing and caching improved query performance significantly.

10. Conclusion

Performance testing proved that the Lease Management System is stable, scalable, and optimized for real-world usage. It efficiently manages concurrent users, maintains quick response times, and sustains performance under stress. Regular performance monitoring and load testing will continue to ensure that the system remains responsive as the number of users and properties grows.

The overall result indicates that the application meets the expected performance benchmarks and is ready for deployment in a real environment.